HCAMatrixBrowser: Accessing the Human Cell Atlas Data in Bioconductor

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Q: How do I programmatically obtain HCA Expression Data in R? 🤔

A: Query the HCA API manually
OR
Use **HCAMatrixBrowser** 🎉

Q: How do I obtain HCA bundle identifiers?

A: From the HCA Data Portal or from HCAExplorer / HCABrowser

Q: How do I filter to certain experimental conditions such as tissue type?

A: Use the available_filters function and the filter function (right panel)

Access the HCA expression matrix data service using **HCAMatrixBrowser**

BiocManager::install("HCAMatrixBrowser")
browseVignettes(package = "HCAMatrixBrowser")

Console | Terminal | Jobs

~/MultiAssayWorkshop/

```
> library(HCAMatrixBrowser)
> hca <- HCAMatrix()
> bundle_fqids <-
+     c("ffd3bc7b-8f3b-4f97-aa2a-78f9bac93775.2019-05-14T122736.345000Z",
+     "f69b288c-fabc-4ac8-b50c-7abcae3731bc.2019-05-14T120110.781000Z",
+     "f8ba80a9-71b1-4c15-bcfc-c05a50660898.2019-05-14T122536.545000Z")
> loadHCAMatrix(hca, bundle_fqids)
class: LoomExperiment
dim: 58347 3
metadata(0):
assays(1): matrix
rownames: NULL
rowData names(9): Accession Gene ... genus_species isgene
colnames(3): 3c2180aa-0aa4-411f-98dc-73ef87b447ed
  1cfe9423-21d1-4281-9f9d-3aaa07b8e1e8 a2a2f604-444c-41b1-befa-25cf7461bf74
colData names(38): CellID barcode ...
  specimen_from_organism.provenance.document_id total_umis
rowGraphs(0): NULL
colGraphs(0): NULL
>
```

Bioconductor
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https://data.humancellatlas.org/

Quick Start

```
library(HCAMatrixBrowser)
hca <- HCAMatrix()
bundle_fqids <- c(
     "ffd3bc7b-8f3b-4f97-aa2a-78f9bac93775.2019-05-14T122736.345000Z",
     "f69b288c-fabc-4ac8-b50c-7abcae3731bc.2019-05-14T120110.781000Z",
     "f8ba80a9-71b1-4c15-bcfc-c05a50660898.2019-05-14T122536.545000Z"
)
loadHCAMatrix(hca, bundle_fqids, format = "loom")
```

Main input

- **bundle_fqids** - bundle identifiers each containing data for a single cell
  - Obtained from HCAExplorer / HCABrowser

Formats (data representation)

- **Loom** -- LoomExperiment
- **MTX** -- SingleCellExperiment
- **CSV** -- list of *tibbles* (very slow)

Filtering

Use the filter function to add filters to the API object

```
hca1 <- filter(hca,
    project.project_core.project_short_name ==
      "Single cell transcriptome analysis of human pancreas" &
    genes_detected >= 300)

loadHCAMatrix(hca1, format = "loom")
```

API Website

https://matrix.data.humancellatlas.org/

Related Bioconductor packages:

- HCAExplorer
- HCABrowser

Note. Changes are coming to the HCA Matrix Service (Data Coordination Platform version 2)! We make use of DCP version 1. Updates to come.